QPS TRIAGE · ABACUS · CODEX — EXECUTION SEED 2026-10-12 · FEDERATED DMAIC

# Global Code Purple

A reusable, recursively self-testing execution protocol across repositories, agents, Docker, CI, PR admission and evidence. It builds on QPS TRIAGE, ABACUS, CODEX, KEB and DOW and does not introduce a second system of authority.

NOT LIFTED · 2 of 5 lift criteria
read back 2026-10-18 13:27 UTC
ABACUS #1858 · merged
ABACUS #1859 · merged
ABACUS: 13 open PRs
CODEX: 0 open PRs
97 merges since unblock

THE CENTRAL RULE · SELECT A STEP TO TRACE IT ON THE PULSE GRAPH

01
**Reuse proven evidence**
Define → Measure

02
**Allocate, preconfigure and pretest**
Measure

03
**Identify the first material red**
Analyse

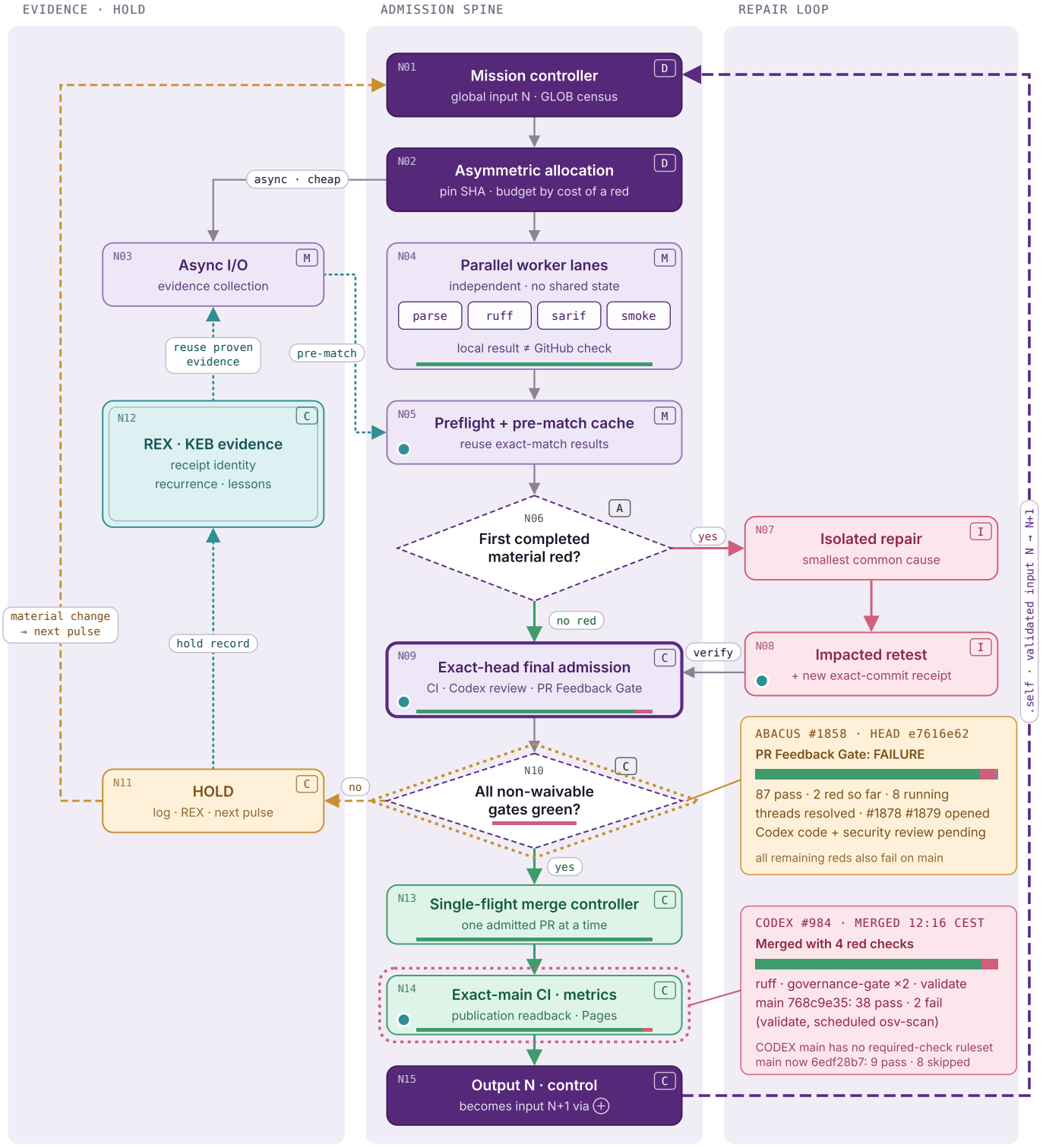
04
**Repair the smallest common cause**
Improve

05
**Independently verify**
Improve

06
**Publish exact-head evidence**
Control

07
**Advance only admissible successors**
Control

## 1 · Execution pulse



## Legend

NODE ROLES

- Authority · resolves or owns state
- Process · checks and verification
- Gate · binary decision
- Repair · isolated change
- Admitted path · merge and proof
- Hold · blocked, logged, retried
- Evidence store · REX and KEB

EDGE TYPES

- Sequence
- Pass
- Material red
- Hold
- Evidence
- .self recursion

MARKERS

- `C` DMAIC phase · hover for card
- `N09` Node ID · hover for card
- Run bar: pass · fail · pending
- Emits an exact-commit receipt
- Where a tracked PR is now

`D Define` `M Measure` `A Analyse` `I Improve` `C Control`

## 2 · Census and telemetry

Iteration census from the `.self` harness: preconfig state per node, run attempts and outcomes, errors, logs, REX lessons, Python runtime and version control. Pass rate counts decisive runs only; pending and neutral runs are shown separately.

| ATTEMPTS | PASSED | FAILED | PENDING | PASS RATE | ERRORS |
|---|---|---|---|---|---|
| 221 | 207 | 13 | 1 | 94% | 5 |
| iteration 0 | decisive green | decisive red | 1 neutral | of decisive runs | 0 REX lessons |

### Coverage · breadth × depth

local run with the CI command on #1858 head 203d63af · 1,086 tests

| OVERALL | LINES | BREADTH | DEPTH | UNTESTED FILES | GATE |
|---|---|---|---|---|---|
| 36.9% | 38.2% | 64.5% | 59.2% | 114 | 75% |
| line + branch | 10,686 / 27,998 | 155 of 269 files touched | inside touched files | 57% of the gap | fixed; red on every PR |

| WAVE | SCOPE | LINE COVERAGE AFTER | TARGET DATE (PROPOSED) |
|---|---|---|---|
| W0 | Floor = measured; new and changed lines ≥ 80% (diff coverage) | 38.2% | 16 Oct |
| W1 | Written disposition for 34 one-shot or legacy files (keep+test, archive, omit) | 46.5% | 20 Oct |
| W2 | DMAIC_V3 phase modules to 80% | 52.1% | 27 Oct |
| W3 | 53 untested DMAIC_V3 modules to 70% | 65.5% | 6 Nov |
| W4 | 34 partly covered DMAIC_V3 modules to 75% | 72.1% | 13 Nov |
| W5 | tools, scripts, models, src, renderers, integration to 80% | 80.2% | 20 Nov |

### Version control

| | |
|---|---|
| REPO | GBOGEB/CODEX · GBOGEB/ABACUS |
| BRANCH | #984 merged · #1858 fix/gate-owner-attestation-fallback |
| HEAD | ABACUS #1858 203d63aff64b9d6b4c69dec0a1d3f5cf3f1f8bf1 |
| BASE | CODEX main 6edf28b7 · ABACUS main 53688f26 (#1838, 2026-10-06) |
| DIRTY TREE | not reported |
| GENERATED | 2026-10-12T15:16:00Z (GitHub API readback) |

### Runtime and debug

| | |
|---|---|
| PYTHON | 3.10 · 3.11 · 3.12 (ABACUS CI matrix, all green on #1858) |
| IMPLEMENTATION | CPython (GitHub-hosted runners) |
| PLATFORM | ubuntu-24.04 |
| CONTAINER | not reported |
| DEBUG DEFAULT | CODEX main: LLDB, lldb-dap + MCP, debugpy + js-debug lanes green |
| PER-NODE | not reported |

## Node census

| ID | NODE | STAGE | TEST | DEBUG | ATT | ✓ | ✗ | PEND | PASS | RUNS | LAST ERROR |
|---|---|---|---|---|---|---|---|---|---|---|---|
| N01 | Mission controller | pass | none | – | — | — | — | — | — | | — |
| N02 | Asymmetric allocation | – | – | – | — | — | — | — | — | | — |
| N03 | Async I/O | – | – | – | — | — | — | — | — | | — |
| N04 | Parallel worker lanes | pass | pass | on | 15 | 15 | 0 | 0 | 100% | | — |
| N05 | Preflight + pre-match cache | – | – | – | — | — | — | — | — | | — |
| N06 | First completed material red? | – | – | – | — | — | — | — | — | | — |
| N07 | Isolated repair | – | – | – | — | — | — | — | — | | — |
| N08 | Impacted retest | – | – | – | — | — | — | — | — | | — |
| N09 | Exact-head final admission | pass | pass | – | 155 | 144 | 10 | 1 | 94% | | PUSH-203d63af New head (workflow_policy.json +1): Code Coverage, Code Quality, Full Red Proof, Security Scanning, policy-and-inventory red |
| N10 | All non-waivable gates green? | – | – | – | 1 | 0 | 1 | 0 | 0% | | NO-GATE CODEX main has no required-check ruleset; #984 merged without a green admission |
| N11 | HOLD | – | – | – | — | — | — | — | — | | — |
| N12 | REX · KEB evidence | – | – | – | — | — | — | — | — | | — |
| N13 | Single-flight merge controller | – | – | – | 1 | 1 | 0 | 0 | 100% | | — |
| N14 | Exact-main CI · metrics | – | – | – | 49 | 47 | 2 | 0 | 96% | | MAIN-768c9e35 validate and scheduled osv-scan failed after the #984 merge; not rerun on later heads |
| N15 | Output N · control | – | – | – | — | — | — | — | — | | — |

### Logs

| | | | |
|---|---|---|---|
| 10-12 17:01 | INFO | N02 | Pinned heads: ABACUS #1858 10fdaaa5; CODEX main 6edf28b7 |
| 10-12 17:01 | WARN | N11 | Nothing merged to ABACUS main since #1838 (2026-10-06) |
| 10-12 16:56 | ERROR | N09 | ABACUS #1858 PR Feedback Gate: state=failure, ci_seen=15 |
| 10-12 16:56 | WARN | N10 | ABACUS gate runs main's code; #1858's own fallback cannot admit #1858 (no self-approval) |
| 10-12 13:11 | INFO | N09 | ABACUS #1858: 3 duplicate gate evaluations cancelled, 1 succeeded |
| 10-12 13:10 | WARN | N11 | #1858 threads L599 and outage correlation: owner residual-risk replies, unresolved |
| 10-12 12:16 | INFO | N01 | Code Purple execution seed merged to CODEX main via #984 (12 commits) |
| 10-12 12:16 | WARN | N10 | CODEX #984 merged with red checks: admission step bypassed |
| 10-12 12:09 | ERROR | N09 | CODEX #984: 4 failing checks on head 90e3396a |

### Errors

| TIME | NODE | CODE | MESSAGE | REX |
|---|---|---|---|---|
| 10-12 12:09 | N09 | CODEX-984 | ruff, governance-gate ×2 and validate failed on merge head 90e3396a | – |
| 10-12 16:56 | N09 | GATE-1858 | PR Feedback Gate failure: 2 unresolved threads; Codex code and security review pending | – |
| 10-12 17:14 | N09 | PUSH-203d63af | New head (workflow_policy.json +1): Code Coverage, Code Quality, Full Red Proof, Security Scanning, policy-and-inventory red | – |
| 10-12 12:16 | N10 | NO-GATE | CODEX main has no required-check ruleset; #984 merged without a green admission | – |
| 10-12 12:20 | N14 | MAIN-768c9e35 | validate and scheduled osv-scan failed after the #984 merge; not rerun on later heads | – |

### REX lessons

No REX lessons captured yet. Capture starts once controls execute.

▼ **Census JSON contract · code-purple.census/v1**

The `.self` harness writes one file per iteration in this shape. Load it with "Load census JSON" to replace the recorded data on every view.

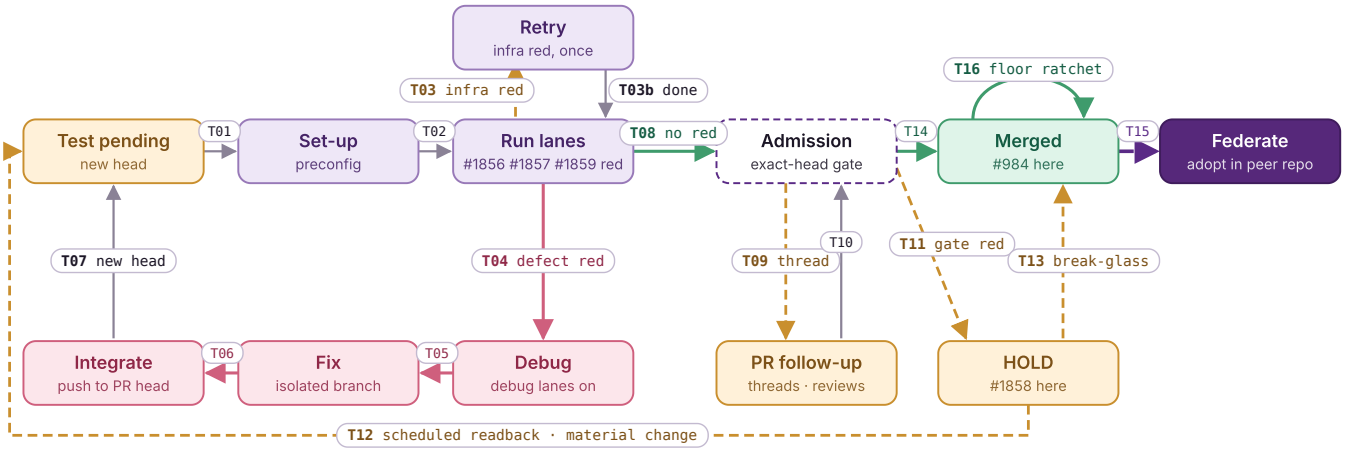
```
{
  "schema": "code-purple.census/v1",
  "iteration": 1,
  "generated_at": "2026-10-16T09:00:00+02:00",
  "vcs": {
    "repo": "GBOGEB/CODEX",
    "branch": "<branch>",
    "head": "<40-char sha>",
    "base": "main@<sha>",
    "dirty": false
  },
  "runtime": {
    "python": "3.11.9",
    "implementation": "CPython",
    "platform": "linux-x86_64",
    "container": "<image or devcontainer>",
    "debug": "off | on | trace"
  },
  "rex": [
    {
      "id": "REX-…",
      "node": "N04",
      "title": "<reusable lesson>",
      "recurrence": 1
    }
  ],
  "nodes": {
    "N04": {
      "preconfig": {
        "stage": "pass | fail | pending | none",
        "test": "pass | fail | pending | none",
        "debug": "off | on | trace"
      },
      "runtime": {
        "python": "3.12.4"
      },
      "runs": [
        {
          "name": "ruff",
          "attempts": 3,
          "pass": 2,
          "fail": 1,
          "neutral": 0,
          "pending": 0,
          "ms": 1800
        }
      ],
      "errors": [
        {
          "ts": "<iso>",
          "code": "RUF100",
          "msg": "<message>",
          "rex": "REX-…"
        }
      ],
      "logs": [
        {
          "ts": "<iso>",
          "level": "ERROR | WARN | INFO | DEBUG",
          "msg": "<message>"
        }
      ]
    }
  }
}
```

## 3 · Process reference

The pulse as a written procedure. Stages run in order; each lists entry criteria, actions and exit outputs, with the nodes it touches. The node and edge tables are generated from the same model as the graph.

### Lifecycle triggers

What moves a work item (a PR head or a red finding) from one state to the next. Each transition has one triggering event, a guard that must hold, and an action that writes evidence. Today only T01's CI start is wired, because GitHub runs checks on pull-request events; the rest are proposed and need the `.self` harness and a readback workflow.



| ID | FROM → TO | TRIGGER EVENT | GUARD | ACTION AND EVIDENCE | NODES | WIRED TODAY |
|---|---|---|---|---|---|---|
| T01 | Test pending → Set-up | Pull request opened or synchronised, or a push | Head SHA pinned | Start CI; mark the gate pending | N01 N02 | Partly wired GitHub starts CI on these events. ABACUS's gate re-evaluates on workflow_run and review events, not on synchronize (#1858 thread, line 599). |
| T02 | Set-up → Run lanes | Preconfig receipt written | Python runtime, dependencies and devcontainer resolved | Start the parallel lanes | N03 N04 | Proposed Needs the .self harness. |
| T03 | Run lanes → Retry | A lane goes red with class runner or infrastructure (5xx, runner lost, cancelled) | No earlier retry on this SHA | Re-run the failed jobs once; log a REX entry | N05 N06 | Manual Re-run all jobs button on GitHub. |
| T03b | Retry → Run lanes | Retry finished | — | Results re-enter triage | N06 | Manual Same re-run. |
| T04 | Run lanes → Debug | Red classified as a source defect that doesn't reproduce locally | It is the first completed material red | Re-run that lane with debug on (debugpy, LLDB) and keep the trace | N06 | Partly wired CODEX main already runs LLDB, lldb-dap and debugpy lanes (green); nothing triggers them from a red yet. |
| T05 | Debug → Fix | Root cause recorded | Fix scope is the smallest common cause | Branch from the exact head | N07 | Manual — |
| T06 | Fix → Integrate | Impacted retest green, with an exact-commit receipt | No governance change approves itself | Push to the PR head | N08 | Manual — |
| T07 | Integrate → Test pending | New head on the PR | — | Old receipts retired; the cycle restarts at T01 | N01 | Wired Push event. |
| T08 | Run lanes → Admission | All lanes terminal | No material red | Exact-head CI plus Codex code and security review | N09 | Partly wired CI is automatic. The Codex review request is manual, and the allowance is exhausted. |
| T09 | Admission → PR follow-up | Review thread, Codex comment or gate failure | — | Answer, fix, or record an owner residual-risk decision | N09 N11 | Partly wired Detection is wired: pr-feedback-gate.yml runs on pull_request_review and review-comment events. The response is manual. |
| T10 | PR follow-up → Admission | Thread resolved or review submitted | Zero unresolved threads | The gate re-evaluates | N10 | Wired Review events trigger the gate. |
| T11 | Admission → HOLD | A non-waivable gate is red or pending (e.g. Codex outage) | — | Write the HOLD record and a REX entry | N10 N11 N12 | Partly wired The gate holds. The REX record is manual. |
| T12 | HOLD → Test pending | Scheduled readback, or a material change | Something changed since the last pulse | Start a new pulse | N11 N01 | Proposed Needs a scheduled task (burndown item 7). |
| T13 | HOLD → Merged | Owner decision with a written residual risk | Owner only. One-time ruleset bypass, restored straight after the merge | Squash-merge, restore the ruleset, write REX | N10 N13 | Manual Owner edits ruleset 24494381. This is the proposed path for #1858. |
| T14 | Admission → Merged | All non-waivable gates green on the same SHA | Single-flight: no other merge in progress | Squash-merge | N13 | Partly wired ABACUS: ruleset enforces it. CODEX: no ruleset; #984 merged with 4 reds. |
| T15 | Merged → Federate | Exact-main CI green and read back | Consumer listed in MESH_DOCK | Open an adoption PR in the peer repo; write the census row | N14 N15 | Proposed Not built. For #984, CODEX main has 2 post-merge reds to clear first. |
| T16 | Merged → Merged | Exact-main CI measures coverage at or above floor + 1 point | Exact-main CI green; measurement on the same SHA | Raise the coverage floor to the measured value; write the census row (total, breadth, depth, per area) | N14 N12 | Proposed Not built. Today the gate is a fixed 75% and fails on every PR. |

## Process stages

### 1 Mobilise

Define

`N01 · Mission controller` `N02 · Asymmetric allocation`

ENTRY
- A trigger: new head, schedule or material change

ACTIONS
- Census authorities and capabilities
- Pin the exact SHA
- Budget lanes by the cost of a red

EXIT
- Pinned SHA pair
- Lane budget

### 2 Collect and run

Measure

`N03 · Async I/O` `N04 · Parallel worker lanes`

ENTRY
- SHA pinned, lanes budgeted

ACTIONS
- Collect prior evidence asynchronously from REX, KEB and GitHub
- Run parse, Ruff, SARIF and smoke lanes in parallel

EXIT
- Evidence bundle
- Per-lane results

### 3 Pre-match

Measure

`N05 · Preflight + pre-match cache`

ENTRY
- Lane results and evidence bundle

ACTIONS
- Reuse results proven for the identical commit and inputs
- Run only the remainder

EXIT
- Preflight receipt (local, not CI proof)

### 4 Triage

Analyse

`N06 · First completed material red?`

ENTRY
- Preflight receipt

ACTIONS
- Rank findings with DOW
- Assign one of four failure classes
- Select the first completed material red

EXIT
- One red to repair, or none

### 5 Repair loop

Improve

`N07 · Isolated repair` `N08 · Impacted retest`

ENTRY
- One classified red

ACTIONS
- Fix the smallest common cause on an isolated branch
- Retest the impacted set
- Write a new exact-commit receipt

EXIT
- Repaired head submitted for verification

### 6 Admission

Control

`N09 · Exact-head final admission` `N10 · All non-waivable gates green?`

ENTRY
- Exact head with no known material red

ACTIONS
- Run CI, Codex review and PR Feedback Gate on the exact SHA
- Evaluate non-waivable gates only

EXIT
- Admit, or hold

### 7 Hold and learn

Control

`N11 · HOLD` `N12 · REX · KEB evidence`

ENTRY
- A non-waivable gate red or pending

ACTIONS
- Record the reason in REX
- Update recurrence and lessons
- Plan the next pulse

EXIT
- Hold record
- Next pulse on material change

### 8 Merge and prove

Control

`N13 · Single-flight merge controller` `N14 · Exact-main CI · metrics` `N15 · Output N · control`

ENTRY
- Admission green

ACTIONS
- Merge one PR at a time
- Run exact-main CI, record metrics, read back the publication
- Write the census row

EXIT
- Output N becomes input N+1

## Node reference

15 of 15 nodes

| ID | NODE | TYPE | PHASE | LANE | IN | OUT | ARTEFACT | EVIDENCE | STATUS |
|---|---|---|---|---|---|---|---|---|---|
| N01 | **Mission controller** Takes global input N, discovers authorities and capabilities, and resolves producer, consumer, receipts and scope for this pulse. | Authority | Define | Admission spine | N11 N15 | N02 | CODEX GLOB.yaml, .mesh/MESH_DOCK.yaml | Census snapshot | Seeded / commi |
| N02 | **Asymmetric allocation** Pins the exact head under test and spends compute where a red is most expensive. Cheap I/O runs async; costly checks get dedicated lanes. | Authority | Define | Admission spine | N01 | N04 N03 | Seed YAML execution contract | Pinned SHA pair, lane budget | Seeded / commi |
| N03 | **Async I/O** Collects existing evidence without blocking the workers, including prior receipts that may already prove a result. | Process | Measure | Evidence · hold | N02 N12 | N05 | REX and KEB lookups, GitHub check readback | Evidence bundle | Not implemente |
| N04 | **Parallel worker lanes** Runs parse, Ruff, SARIF security scans and smoke in parallel with no shared state. | Process | Measure | Admission spine | N02 | N05 | ABACUS scripts/testpilot_preflight.py, pre-commit, Ruff, Bandit, Semgrep | Per-lane JSON and SARIF | Partial |
| N05 | **Preflight + pre-match cache** Merges worker output with collected evidence. A result already proven for the identical commit and inputs is reused instead of rerun. | Process | Measure | Admission spine | N04 N03 | N06 | Receipts keyed by exact commit and input hash | Preflight receipt | Not implemente |
| N06 | **First completed material red?** Ranks findings and selects the first completed red that matters, classified as source defect, inherited debt, review capacity or runner failure. | Gate | Analyse | Admission spine | N05 | N07 N09 | ABACUS DOW | Ranked disposition | Not implemente |
| N07 | **Isolated repair** Fixes the smallest common cause only. Security-sensitive governance changes cannot approve themselves. | Repair | Improve | Repair loop | N06 | N08 | Smallest-scope branch | Commit | Not implemente |
| N08 | **Impacted retest** Reruns only what the repair touches and writes a new exact-commit receipt. | Repair | Improve | Repair loop | N07 | N09 | Impacted test set | Exact-commit receipt | Not implemente |
| N09 | **Exact-head final admission** Independent verification on the exact head SHA, including required reviews. | Process | Control | Admission spine | N08 N06 | N10 | GitHub Actions, PR Feedback Gate, Codex review and security review | Check records, review states | In progress |
| N10 | **All non-waivable gates green?** Admits only when every non-waivable check and review is green on the same SHA. Waivable checks are recorded, not counted. | Gate | Control | Admission spine | N09 | N11 N13 | Required checks and reviews at exact head | Admission decision | On hold |
| N11 | **HOLD** Logs the blocking reason to REX and plans the next pulse. A material change re-enters at the mission controller. | Hold | Control | Evidence · hold | N10 | N12 N01 | REX entry | Hold record | On hold |
| N12 | **REX · KEB evidence** Holds evidence identity, recurrence counts and reusable lessons. Async I/O reads from it at the start of each pulse. | Evidence store | Control | Evidence · hold | N11 | N03 | CODEX KEB and REX | Lessons, recurrence | Partial |
| N13 | **Single-flight merge controller** Merges one admitted PR at a time so no head goes stale behind another merge. | Admitted path | Control | Admission spine | N10 | N14 | Merge queue discipline | Merge commit | Not reached |
| N14 | **Exact-main CI · metrics** Proves main is green after the merge, records metrics and reads the published result back. | Admitted path | Control | Admission spine | N13 | N15 | GitHub Actions on main, Pages | Post-merge receipt | Not reached |
| N15 | **Output N · control** Becomes the input of iteration N+1 through governed reconciliation (⊕). | Authority | Control | Admission spine | N14 | N01 | Iteration census | Census row | Not reached |

## Edge reference

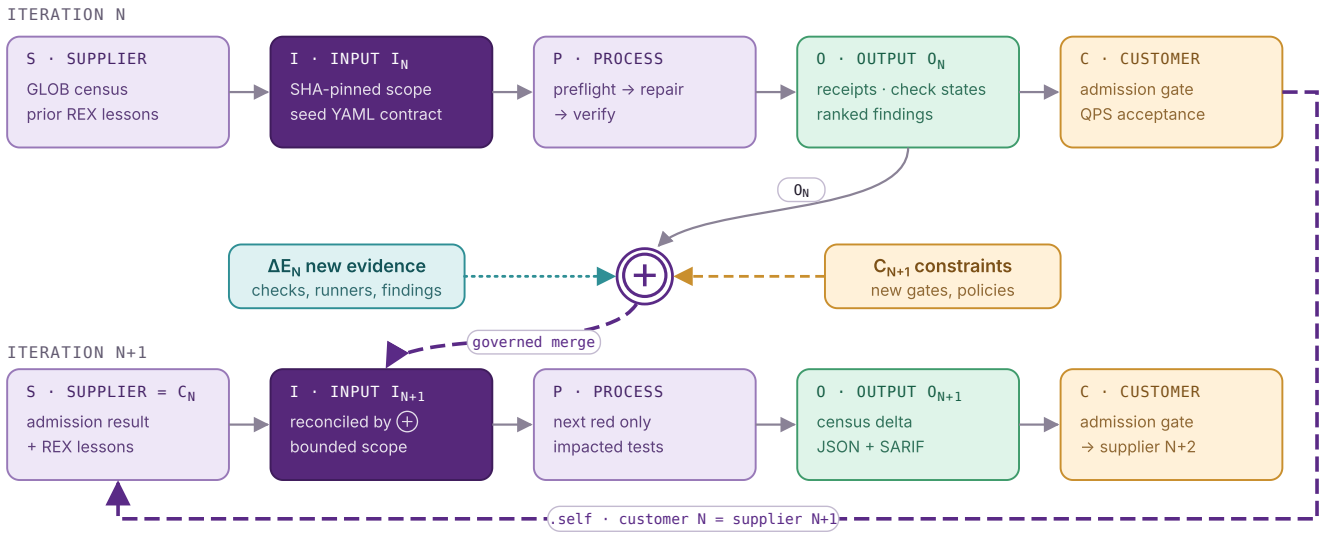
| ID | FROM | TO | TYPE | LABEL | MEANING |
|---|---|---|---|---|---|
| E01 | N01 Mission controller | N02 Asymmetric allocation | Sequence | — | Every pulse starts at the controller. |
| E02 | N02 Asymmetric allocation | N04 Parallel worker lanes | Sequence | — | Costly checks get dedicated lanes. |
| E03 | N02 Asymmetric allocation | N03 Async I/O | Sequence | `async · cheap` | Cheap I/O runs without blocking the workers. |
| E04 | N04 Parallel worker lanes | N05 Preflight + pre-match cache | Sequence | — | Per-lane JSON and SARIF results. |
| E05 | N03 Async I/O | N05 Preflight + pre-match cache | Evidence | `pre-match` | Prior receipts offered for exact-match reuse. |
| E06 | N05 Preflight + pre-match cache | N06 First completed material red? | Sequence | — | Merged preflight result. |
| E07 | N06 First completed material red? | N07 Isolated repair | Material red | `yes` | A completed material red exists. |
| E08 | N07 Isolated repair | N08 Impacted retest | Material red | — | Repair committed on an isolated branch. |
| E09 | N08 Impacted retest | N09 Exact-head final admission | Sequence | `verify` | Retest receipt goes to exact-head verification. |
| E10 | N06 First completed material red? | N09 Exact-head final admission | Pass | `no red` | No material red among completed checks. |
| E11 | N09 Exact-head final admission | N10 All non-waivable gates green? | Sequence | — | Check and review states on the exact head. |
| E12 | N10 All non-waivable gates green? | N11 HOLD | Hold | `no` | A non-waivable gate is red or pending. |
| E13 | N10 All non-waivable gates green? | N13 Single-flight merge controller | Pass | `yes` | Every non-waivable gate is green on the same SHA. |
| E14 | N13 Single-flight merge controller | N14 Exact-main CI · metrics | Pass | — | One PR merged at a time. |
| E15 | N14 Exact-main CI · metrics | N15 Output N · control | Pass | — | Main green; metrics and publication read back. |
| E16 | N11 HOLD | N12 REX · KEB evidence | Evidence | `hold record` | Blocking reason recorded. |
| E17 | N12 REX · KEB evidence | N03 Async I/O | Evidence | `reuse proven evidence` | Lessons and receipts reused in the next pulse. |
| E18 | N11 HOLD | N01 Mission controller | Hold | `material change → next pulse` | A material change triggers the next pulse. |
| E19 | N15 Output N · control | N01 Mission controller | .self recursion | `.self · validated input N → N+1` | Validated output becomes input N+1 through ⊕. |

## Mermaid export

```
flowchart TD
  subgraph L_spine["Admission spine"]
    N01["N01 · Mission controller"]:::auth
    N02["N02 · Asymmetric allocation"]:::auth
    N04["N04 · Parallel worker lanes"]:::proc
    N05["N05 · Preflight + pre-match cache"]:::proc
    N06{"N06 · First completed material red?"}:::dec
    N09["N09 · Exact-head final admission"]:::proc
    N10{"N10 · All non-waivable gates green?"}:::dec
    N13["N13 · Single-flight merge controller"]:::pass
    N14["N14 · Exact-main CI · metrics"]:::pass
    N15(["N15 · Output N · control"]):::auth
  end
  subgraph L_ev["Evidence · hold"]
    N03["N03 · Async I/O"]:::proc
    N11["N11 · HOLD"]:::hold
    N12[("N12 · REX · KEB evidence")]:::evid
  end
  subgraph L_rep["Repair loop"]
    N07["N07 · Isolated repair"]:::rep
    N08["N08 · Impacted retest"]:::rep
  end
  N01 --> N02
  N02 --> N04
  N02 -->|"async · cheap"| N03
  N04 --> N05
  N03 -.->|"pre-match"| N05
  N05 --> N06
  N06 -->|"yes"| N07
  N07 --> N08
  N08 -->|"verify"| N09
  N06 -->|"no red"| N09
  N09 --> N10
  N10 -.->|"no"| N11
  N10 -->|"yes"| N13
  N13 --> N14
  N14 --> N15
  N11 -.->|"hold record"| N12
  N12 -.->|"reuse proven evidence"| N03
  N11 -.->|"material change → next pulse"| N01
  N15 ==>|".self · validated input N → N+1"| N01
  classDef auth fill:#56287a,color:#fff,stroke:#3f1c5c
  classDef proc fill:#eee7f7,color:#452366,stroke:#9879b8
  classDef dec fill:#fff,color:#1c1626,stroke:#5a2a86,stroke-dasharray:5 3
  classDef rep fill:#fce6eb,color:#8f2747,stroke:#cf5f7d
  classDef pass fill:#dff4e9,color:#145c43,stroke:#3f9b6c
  classDef hold fill:#fff1d9,color:#7c5118,stroke:#c98f2e
  classDef evid fill:#def1f2,color:#135458,stroke:#2f8f95
  linkStyle 9,12,13,14 stroke:#3f9b6c,stroke-width:2px
  linkStyle 6,7 stroke:#cf5f7d,stroke-width:2px
  linkStyle 11,17 stroke:#c98f2e,stroke-width:2px,stroke-dasharray:6 4
  linkStyle 4,15,16 stroke:#2f8f95,stroke-dasharray:2 4
  linkStyle 18 stroke:#5a2a86,stroke-width:3px
```

## 4 · Recursive SIPOC

A standard SIPOC runs once, left to right. With `.self`, the customer of iteration N becomes the supplier of iteration N+1, and the next input is a governed reconciliation, not a concatenation.




### State transition

$$I_{N+1} = O_N \oplus \Delta E_N \oplus C_{N+1}$$

| | |
|---|---|
| $O_N$ | Last output: receipts, check states, ranked findings. |
| $\Delta E_N$ | Newly observed evidence: completed checks, runner behaviour, new findings. |
| $C_{N+1}$ | Newly applicable constraints: gates, policies, review rules. |
| $\oplus$ | Governed reconciliation. Superseded evidence is retired, conflicts go to HOLD, nothing is appended blindly. |

### Confidence is not admission

A passing local preflight raises confidence for the next iteration. Only exact-head GitHub checks and reviews move a PR through the admission gate.

## 5 · Lanes and dock

Ten execution lanes grouped by what they contribute. Each lane keeps its existing canonical owner; Code Purple only routes between them. "Show on pulse" traces the lane onto the execution graph.

PROTOCOL
**Code Purple**

### TIER 1 — Discover

#### Global lookup

`GLOB.yaml · global mesh census`

Discover authorities and available capabilities.

`CODEX`

#### Dock interface

`.mesh/MESH_DOCK.yaml`

Resolve producer, consumer, receipts and scope.

`mesh`

### TIER 2 — Verify

#### Runtime and self-test

`scripts/testpilot_preflight.py`

Fast preflight, smoke and testflight.

`ABACUS`

#### Lint and configuration

`pre-commit · Ruff · YAML/JSON/TOML`

Detect deterministic defects early.

`both`

#### Coverage

`ci-cd-tests.yml · MIP B0 coverage evidence`

Line + branch per test; breadth, depth and per-area census; floor ratchet (T16).

`ABACUS`

#### Security

`Bandit · Semgrep · SARIF`

Typed findings. No fabricated security PASS.

`both`

### TIER 3 — Analyse and remember

#### Derived analysis

`DOW`

Rank findings and return scoped dispositions.

`ABACUS`

#### Semantic and provenance

`KEB · REX`

Evidence identity, recurrence and reusable lessons.

`CODEX`

### TIER 4 — Authorise and deliver

#### QPS domain

`cryoplant-project controlled sources`

Engineering authority over acceptance stays here.

`cryoplant-project`

#### Delivery

`GitHub Actions · PR Feedback Gate · Pages`

Exact-head admission and post-merge proof.

`GitHub`

### Two meanings of "dock"

`logical` The mesh contract in `MESH_DOCK.yaml`: who produces, who consumes, which receipts count, what scope applies.

`runtime` Container networking: ports a devcontainer or compose service exposes. Configuration alone proves nothing about a running service.

### Runtime dock status ladder

| SURFACE | CONFIGURED | LISTENING | PUBLISHED | PROBED |
|---|---|---|---|---|
| ABACUS devcontainer<br>`8000 · 8080 · 5000 forwarded` | ● | ◌ | ◌ | ◌ |
| CODEX compose validator<br>`no published ports` | ● | ◌ | — | ◌ |

● verified ◌ unproven — not declared

## 6 · Admission board

The inbox shows an execution bottleneck rather than a coding backlog. Each red gets one of four causes before anyone repairs it, so a runner outage is never fixed as a source defect.

RED IN SCOPE
**4**
ABACUS #1856 #1857 #1858 #1859

READY TO MERGE
**0**
ABACUS main idle since #1838 (10-06)

OPEN ABACUS PRS
**20**
all held by the Codex review gate

MERGED TODAY
**1**
CODEX #984, with 4 red checks

### Failure classes

**Source defect**
Introduced by this PR. Repair in the PR.

**Inherited quality debt**
Already on main. Ratchet new debt; keep full-main enforcement.

**Review capacity**
Codex review or security review unavailable. Owner fallback, never self-approval.

**Runner or infrastructure**
Not a code fault. Retry or fix the runner, then re-read.

### Pull requests in scope · read back 17:55 CEST

ABACUS #1858 — P0 · gate FAILURE

**Owner-attested review fallback**

Head e7616e62 (17:52): regenerated rationalization doc, policy-and-inventory green. Remaining reds (Code Quality, Security Scanning; Coverage and Full Red Proof still running at 17:55) all fail nightly on ABACUS main: inherited debt. Threads resolved; the follow-up fixes are already in the PR (a7f623b5, 14430046). Gate pending on Codex reviews, so landing needs the owner break-glass.

ABACUS #1859 — 4 red

**PR-level new-debt ratchet**

Head 6fac58cc. Red: Code Quality Checks, Full Red Proof Summary, Security Scanning, validate-scripts. Class not yet assigned. Distinct control from #1858; must not weaken full-main enforcement.

ABACUS #1857 — 4 red

**Fix syntax error in diagnostic_kpi_probe.py**

Head e7b45912. Red: Code Coverage, Code Quality Checks, Full Red Proof Summary, Security Scanning. Likely inherited debt (the same reds as #1856); class not yet confirmed.

ABACUS #1856 — 3 red

**Lossless branch hygiene burn-down**

Head f11d8044. Red: Code Coverage, Code Quality Checks, Full Red Proof Summary. Same reds recur across PRs, which points to inherited debt; #1859 targets exactly this.

CODEX #984 — merged 12:16

**Global Code Purple seed**

Merged with 4 red checks on head 90e3396a (ruff, governance-gate ×2, validate). CODEX main has no required-check ruleset, so nothing stopped it. Post-merge 768c9e35: 38 pass, 2 fail (validate, scheduled osv-scan). Current main 6edf28b7 green.

## Burndown

| # | ITEM | STATUS | NEXT ACTION | OWNER | TARGET (PROPOSED) |
|---|---|---|---|---|---|
| 1 | Land ABACUS #1858 | in progress | Done: threads resolved (17:20), #1878/#1879 opened, rationalization doc pushed (e7616e62, policy-and-inventory green). Next: owner accepts the 4 inherited reds and adds the one-time bypass; Claude squash-merges and reads back; owner restores the ruleset. | GBO (ruleset, merge) · Claude (threads, issues, readback) | 2026-10-13 |
| 2 | Follow-up hardening for #1858 | opened | #1878 gate pending on `pull_request: synchronize`; #1879 correlate outage reply with a current-head request. | Open for implementation | 2026-10-13 |
| 3 | CODEX post-merge reds | classified | validate fixed on main at f9ae4676. OSV-Scanner fails on every main push since before #984: inherited, needs its own fix. | Claude | 2026-10-13 |
| 4 | CODEX admission gap | decision | Decide whether CODEX main gets a required-check ruleset like ABACUS. | GBO | 2026-10-14 |
| 5 | Classify reds on #1856, #1857, #1859 | confirmed | Code Coverage, Code Quality, Full Red Proof and Security Scanning fail in every nightly run on ABACUS main since at least 7 Oct: inherited debt. | Claude | 2026-10-14 |
| 6 | Runnable `.self` harness | todo | Emit census JSON (contract in the Census tab), SARIF findings and exact-commit receipts. | Claude | 2026-10-16 |
| 7 | Scheduled readback pulse (T12) | todo | Read back tracked PRs and both mains on a schedule; update this page's data. | Claude | 2026-10-16 |
| 8 | Docker smoke harness, live port probing | not built | After the harness contract is stable. | Claude | 2026-10-23 |
| 9 | Cross-repository adoption (T09) | gated | Adopt the seed in ABACUS once #1858 has landed. | — | gate: item 1 |
| – | QPS completed lanes | guard | Preserve; no regression from Code Purple changes. | — | ongoing |

## 7 · Lift and method

**Not lifted.** Merge flow returned when #1858 merged on 12 October, but the ruleset was emptied during the break-glass and never restored: since then no merge has passed a green PR Feedback Gate (EVERRED root EVR-ABACUS-0001). These criteria are read from GitHub by `kit/readback.py`; the eight steps below are the reusable method, adapted per incident through one file, `kit/incident.yaml`.

OPEN ABACUS PRS
**20 → 2**
declaration → 13 Oct 17:43

MERGES IN 23 H
**44**
after 6 days with none

COVERAGE
**40.3%**
from 36.9%; 1,247 tests

RULESET RULES
**0**
emptied 12 Oct 17:58; main unprotected

| ID | LIFT CRITERION | MET | EVIDENCE |
|---|---|---|---|
| L1 | Unblocker PR merged | yes | #1858 merged 2026-10-12 18:17 CEST (e10e39d3) |
| L2 | Break-glass bypass removed | yes | Ruleset 24494381 active, 0 bypass actors |
| L3 | Merge flow restored (≥ 5 merges since the unblocker) | yes | 43 merges after #1858 |
| L4 | Backlog ≤ 25% of baseline | yes | 2 open (baseline 20) |

### Worked

- Readback before action showed #1858 could not admit itself and #984 had merged red.
- Classifying against main's nightly run proved four reds inherited.
- The smallest fix (a 3-line regenerated doc) cleared the only red the PR caused.
- Owner/agent split: owner bypass, merge and restore; agent threads, issues, fix, readback.

### Change next time

- Run the readback every pulse and on a schedule; snapshots went stale twice in an hour.
- Re-read the PR head before writing a decision on it.
- Decide on a required-check ruleset for CODEX (#984 merged with 4 reds).
- Break glass with a bypass actor, never by deleting rules; the lift check now compares actual rules to expected rules (it passed falsely on 13 Oct).
- Burn down the four inherited reds; coverage W0 due 16 Oct.

## The method in eight steps

| STEP | NAME | ENTRY | ACTION | WHO |
|---|---|---|---|---|
| 0 | **Declare** | Merge flow stalled, the same checks red across PRs, or a required reviewer unavailable | Write incident.yaml | Owner |
| 1 | **Read back** | incident.yaml exists | Run readback.py; no action before the first census | Agent |
| 2 | **Classify every red** | Census on exact heads | Inherited · source defect · review capacity · runner or infra | Agent |
| 3 | **Find the unblocker** | Classes known | Smallest change that lets the gate admit again; prove it cannot admit itself | Agent + owner |
| 4 | **Repair source defects only** | Unblocker chosen | Fix, impacted retest, exact-head readback | Agent |
| 5 | **Admit or break glass** | Only inherited or capacity reds left | Owner writes residual risk, one-time bypass, merge, remove bypass | Owner |
| 6 | **Lift check** | Unblocker merged | readback.py evaluates the L-criteria | Agent |
| 7 | **Close out** | Lifted | Verify follow-ups on main and close; REX lesson; inherited reds become dated burn-down items | Agent + owner |

Toolkit files: incident.yaml · readback.py · after_action.md · code-purple.md